THE CYLINDER ON A SIMPLICIAL OBJECT IS CONTRACTIBLE OVER THE OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-contractible (source lines 4830-4844); the proof environment begins two lines below it and is its own proof, closing at source line 4871.

Editorial note. The statement and the proof invoke the author's lemma on the maps between U and its cylinder by label; that lemma's statement and its own proof are retained as a context span, and the generator expands the context transitively, so the reference resolves inside the assembled package and no source pointer is declared.

Editorial note. The proof displays one formula for the homotopy; it is part of the retained source range and is reproduced as written, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: the cylinder on a simplicial object is contractible over the object, with explicit homotopies through each end.

Allowed standard prerequisites (editorial).

- Product of a simplicial object with the standard simplex. The cylinder U times the standard 1-simplex, computed degreewise, together with the two end inclusions and the projection (the author's lemma lemma-back-to-U, retained as context).. It supplies the description of the maps e_0, e_1 and pi used in the statement and the proof (source lines 4834-4836, 4846-4869).
- Homotopy between morphisms of simplicial objects. Two morphisms are homotopic when there is a family of degreewise maps satisfying the face and degeneracy identities of a homotopy.. The proof constructs such families explicitly, using maxima of simplex coordinates for the first and minima for the second (source lines 4850-4870).
- Finite coproducts in the base category. The category admits finite coproducts, which is what allows the standard simplices and their products with U to be formed degreewise.. It is the hypothesis that makes the cylinder and the maps exist (source lines 4832-4833).

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Lemma 0.1. *Let $\mathcal{C}$ be a category such that the coproduct of any two objects of $\mathcal{C}$ exists. Let us temporarily denote FSSets the category of simplicial sets all of whose components are finite nonempty.*

(1) *The rule $(U, V) \mapsto U \times V$ defines a functor FSSets $\times$ Simp($\mathcal{C}$) $\to$ Simp($\mathcal{C}$).*
(2) *For every U, V as above there is a canonical map of simplicial objects*

$$U \times V \longrightarrow V$$

defined by taking the identity on each component of $(U \times V)_n = \coprod_u V_n$.

Proof. Omitted. □

Lemma 0.2. *Let $\mathcal{C}$ be a category with finite coproducts. Let U be a simplicial object of $\mathcal{C}$. Consider the maps $e_1, e_0 : U \to U \times \Delta[1]$, and $\pi : U \times \Delta[1] \to U$, see Lemma 0.1.*

(1) *We have $\pi \circ e_1 = \pi \circ e_0 = id_U$, and*
(2) *The morphisms $id_{U \times \Delta[1]}$, and $e_0 \circ \pi$ are homotopic.*
(3) *The morphisms $id_{U \times \Delta[1]}$, and $e_1 \circ \pi$ are homotopic.*

Proof. The first assertion is trivial. Consider the map of simplicial sets $\Delta[1] \times \Delta[1] \longrightarrow \Delta[1]$ which in degree n assigns to a pair (β_1, β_2), $\beta_i : [n] \to [1]$ the morphism $\beta : [n] \to [1]$ defined by the rule

$$\beta(i) = \max\{\beta_1(i), \beta_2(i)\}.$$

It is a morphism of simplicial sets, because the action $\Delta[1](\varphi) : \Delta[1]_n \to \Delta[1]_m$ of $\varphi : [m] \to [n]$ is by precomposing. Using the notation $\Delta[1]_n = \{\alpha_0^n, \ldots, \alpha_{n+1}^n\}$ introduced above, we have $\beta = \alpha_0^n$ if $\beta_2 = \alpha_0^n$ and $\beta = \beta_1$ if $\beta_2 = \alpha_{n+1}^n$. Since α_0^n, resp. α_{n+1}^n is the constant map with value 1, resp. 0, this implies that the induced morphism

$$U \times \Delta[1] \times \Delta[1] \longrightarrow U \times \Delta[1]$$

of Lemma 0.1 is a homotopy from $\mathrm{id}_{U \times \Delta[1]}$ to $e_1 \circ \pi$. Similarly for $e_0 \circ \pi$ (use minimum instead of maximum). □